

PRODUCT REQUIREMENT DOCUMENTATION (PRD)

Product Name: Learners (FRSC-Verified Driving School & Safety Tracking Platform)

1. Overview

1.1 Purpose

The purpose of Learners is to reduce road accidents in Nigeria by digitizing and standardizing driver education through a verified network of FRSC-approved driving schools, while ensuring real-time safety tracking and accountability during training sessions.

1.2 Problem Statement

Road traffic accidents in Nigeria are significantly driven by:

- Poor-quality driver training
- Unverified driving schools
- Lack of oversight during driving lessons
- No formal record of practical driving experience

Learners and guardians currently have no reliable way to:

- Verify instructor credibility
- Confirm training vehicle safety
- Track driving sessions in real time
- Retain structured driving history

2. Goals & Objectives

Primary Goals

- Improve the quality and accountability of driving education
- Increase learner and guardian confidence in driving training
- Support safer roads through better-trained drivers

Success Metrics (KPIs)

- Number of verified driving schools onboarded
- Number of completed training sessions
- Average instructor rating
- Reduction in reported safety incidents
- User retention (repeat bookings)

3. Target Market

Initial Launch

- **Location:** Lagos State, Nigeria
- **Users:**
 - New drivers (18–35)
 - Parents/guardians enrolling wards
 - FRSC-approved driving schools

Expansion (Phase 2)

- Other metropolitan cities (Abuja, Port Harcourt, Ibadan)
- Nationwide rollout

4. User Personas

4.1 Learner / Customer

- Wants safe, structured driving lessons
- Needs verified instructors and vehicles
- Wants proof of learning progress

4.2 Parent / Guardian

- Wants real-time visibility during lessons
- Needs assurance of safety and accountability

4.3 Driving School / Instructor

- Wants increased visibility and customer acquisition
- Needs scheduling, payment, and performance tracking tools

4.4 Admin

- Oversees platform integrity
- Verifies schools and resolves disputes

5. Scope

In Scope (MVP)

- Driving school onboarding & verification
- Location-based discovery
- Booking & payments
- Vehicle verification
- Real-time lesson tracking
- Training history storage
- Ratings & reviews

Out of Scope (MVP)

- Driving license issuance
- Vehicle insurance underwriting
- Accident claim processing

6. Functional Requirements

6.1 Authentication & User Management

- Email/phone-based registration
- Role-based access:
 - Learner
 - Guardian
 - Driving School
 - Admin
- Profile management

6.2 Driving School Onboarding & Verification

- School submits:
 - FRSC approval number
 - Business address
 - Instructor licenses
- Admin approval required before listing
- Ability to suspend or revoke access

6.3 Discovery & Search

- Map-based search (Lagos-focused)
- Filters:
 - Distance
 - Price
 - Vehicle type (manual/automatic)
 - Training package
- Driving school profile page with:
 - Instructor list
 - Vehicle details
 - Ratings & reviews

6.4 Booking & Scheduling

- View instructor availability
- Book individual or bundled sessions
- Calendar-based scheduling
- Automated booking confirmations

6.5 Vehicle Transparency

Before booking, users must see:

- Vehicle images
- Plate number
- Vehicle type
- Roadworthiness status (manual upload for MVP)
- Insurance details (optional, Phase 2)

6.6 Payments

- In-app payments
- Payment held in escrow
- Instructor payout after session completion
- Transaction history & receipts

6.7 Driving Session Tracking (Core Feature)

Start Session

- Learner or instructor starts session
- System validates:
 - Assigned vehicle
 - Assigned instructor
- GPS tracking begins

Live Tracking

- Guardian can track in real time:
 - Location
 - Route
 - Speed (basic)
- Push notifications for:
 - Session start
 - Session end

End Session

- Session is ended by either Instructor or Learner to avoid further tracking.
- Data stored for historical reference

6.8 Training History & Records

- Driving hours log
- Routes driven
- Session summaries
- Downloadable driving log (Phase 2)

6.9 Ratings & Feedback

- Post-session rating for:
 - Instructor

- Vehicle
- Safety issue reporting
- Admin review & escalation

6.10 Admin Dashboard

- Approve/reject driving schools
- Monitor live sessions
- User management
- Analytics dashboard:
 - Session count
 - Instructor performance
 - Incident reports

7. Non-Functional Requirements

7.1 Performance

- Real-time tracking latency ≤ 5 seconds
- App load time ≤ 3 seconds on 3G networks

7.2 Security

- Encrypted GPS and personal data
- Role-based authorization
- Audit logs for session tracking

7.3 Compliance

- NDPR (Nigeria Data Protection Regulation)
- Consent-based GPS tracking

8. User Journey (High-Level)

1. User registers → verifies identity
2. Searches nearby driving schools
3. Views instructor & vehicle details
4. Books and pays for session
5. Session starts → live tracking begins
6. Session ends → data stored
7. User rates instructor

9. Technical Architecture (High-Level)

Frontend

- React / React Native
- Map integration (Google Maps / Mapbox)

Backend

- NestJS
- PostgreSQL or MongoDB
- WebSockets for real-time tracking

Infrastructure

- Cloud hosting (AWS/GCP)
- Object storage for vehicle images

10. Risks & Mitigations

Risk

Fake driving schools

Privacy concerns

GPS inaccuracies

Instructor misuse

Mitigation

Strict FRSC verification

Explicit tracking consent

Fallback polling

Ratings & monitoring

11. Assumptions & Dependencies

- FRSC verification process is accessible
- Driving schools are willing to adopt digital tools
- Mobile internet is available during lessons

12. Future Enhancements

- FRSC system integration

- AI-based driving behavior analysis
- Digital learner certification
- Insurance & car rental partnerships
- Nationwide expansion

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